Statistical Distributions & Testing

Theory Guide with Definitions

Part A – Theoretical Foundation
Spread Locator Transaction Data Analysis Project

Topic	Used for (in this project)
1. Statistical distributions	Framework for every model below
2. Q-Q plot	Test normality of transaction amounts
3. Discrete vs continuous	Choose the right model for each column
4. Bernoulli	Transaction status (Success/Fail)
5. Binomial	Weekly transaction count
6. Log-Normal	Transaction amount (best fit)
7. Power law	Heavy-tail check for amounts
8. Box-Cox transform	Normalise and stabilise amounts
9. Poisson	Transactions per day
10. Z-score probability	P(amount > ₹5,000), outlier detection
11. PDF vs CDF	Visualise and read amount probabilities
Appendix	Glossary, formula cheat sheet, distribution chooser

How to read this guide: every section follows the same pattern – **Definition → Formula → Key properties → Example → Use in this project**.

1. Statistical Distributions

Definition. A statistical distribution describes **all the values a random variable can take and how likely each value (or range of values) is**. It is specified by a function – the PMF, PDF or CDF – and a small number of parameters.

Think of it as the "shape" of the data: where values cluster, how far they spread, and how fast the probability dies away in the tails.

Key ideas

- **Random variable (X):** a quantity whose value depends on chance, e.g. a transaction amount.
- **Parameters:** numbers that fix the exact shape, e.g. mean μ and standard deviation σ for the Normal; rate λ for the Poisson; probability p for the Bernoulli.
- **Why we fit distributions:** they summarise data compactly, let us compute probabilities, support hypothesis tests and allow simulation and forecasting.
- **Fitting:** choose the parameters that make the model match the data best (maximum likelihood estimation, MLE), then judge the match with plots and tests.

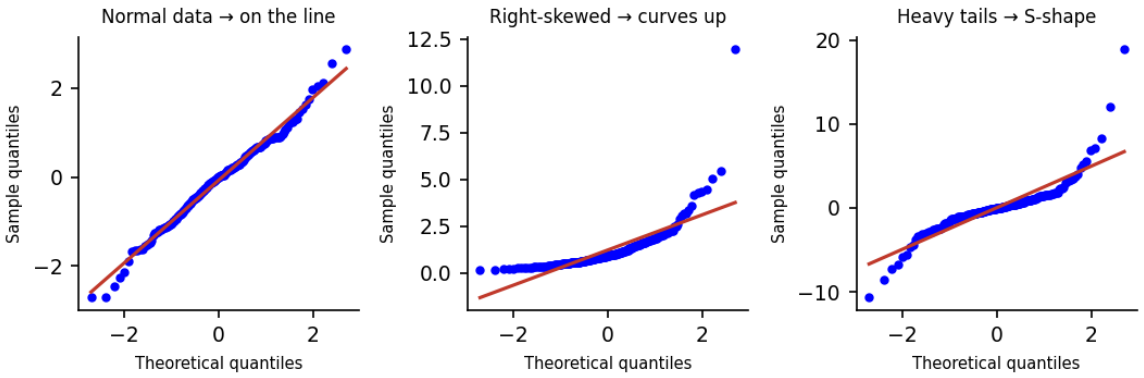
Example. Daily number of transactions averaging 7.1 can be modelled as Poisson($\lambda = 7.1$); the amount of each transaction can be modelled as Log-Normal($\mu = 8.0$, $\sigma = 0.47$).

Use in this project. Part B fits Bernoulli, Binomial, Poisson, Log-Normal and Power law models to the dataset and decides which one describes each column best.

2. Q-Q Plot (Quantile-Quantile Plot)

Definition. A Q-Q plot compares the **quantiles of the observed data** (y-axis) with the **quantiles of a theoretical distribution** (x-axis, usually the Normal). If the data follow that distribution, the points fall along a straight reference line.

A **quantile** is a cut-off below which a given fraction of the data lies (the median is the 50% quantile).



How to read it

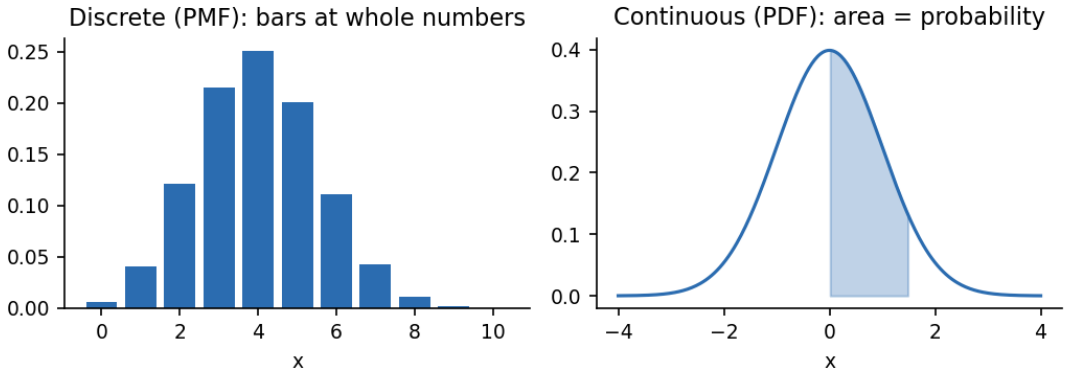
- **Points on the line:** data are approximately normal.
- **Curve bending upwards at the right end:** right-skewed data / long right tail (typical of money amounts).
- **S-shape:** heavy tails (more extreme values than a Normal).
- **Isolated points far from the line:** outliers.

Why it is used. Many methods (t-test, regression, ANOVA, Z-score probabilities) assume normality. A Q-Q plot checks this assumption *visually* and shows *where* the data deviate, which a single test statistic cannot. It is usually paired with a formal test such as Shapiro-Wilk ($p < 0.05 \Rightarrow$ not normal).

Use in this project. The raw transaction amounts curve sharply away from the line, whereas log(amount) lies almost on it – evidence for a Log-Normal model.

3. Discrete vs Continuous Distributions

Definition. A **discrete** distribution is for a variable that takes **countable** values (0, 1, 2, ...). A **continuous** distribution is for a variable that can take **any value in an interval** (e.g. 3,481.27).



Aspect	Discrete	Continuous
Possible values	Countable, separate	Any real value in a range
Described by	Probability Mass Function (PMF)	Probability Density Function (PDF)
P(X = x)	Can be greater than 0	Always 0 – only intervals have probability
Probabilities sum/integrate to 1	Σ P(x) = 1	$\int$ f(x) dx = 1
Plot	Bars	Smooth curve
Examples	Bernoulli, Binomial, Poisson	Normal, Log-Normal, Pareto (Power law)
In this dataset	transaction_status, transaction_count	transaction_amount

Rule of thumb: if you *count* it, it is discrete; if you *measure* it, it is continuous.

4. Bernoulli Distribution

Definition. The Bernoulli distribution models a **single trial with exactly two outcomes**: success (1) with probability p and failure (0) with probability $1 - p$.

$$P(X = x) = p^x (1 - p)^{1-x}, x \in \{0, 1\}$$

Property	Value
Mean	p
Variance	p(1 − p)
Parameter estimate (MLE)	$\hat{p}$ = (number of successes) / n

Example. A coin that lands heads with probability 0.5; a transaction that is successful with probability 0.445.

Use in this project. transaction_status is coded Success = 1, Fail = 0. The fitted p ≈ 0.445, so about 55% of transactions fail. "Transaction occurrence" (did the customer transact at least once that week?) is also Bernoulli, with p ≈ 0.93.

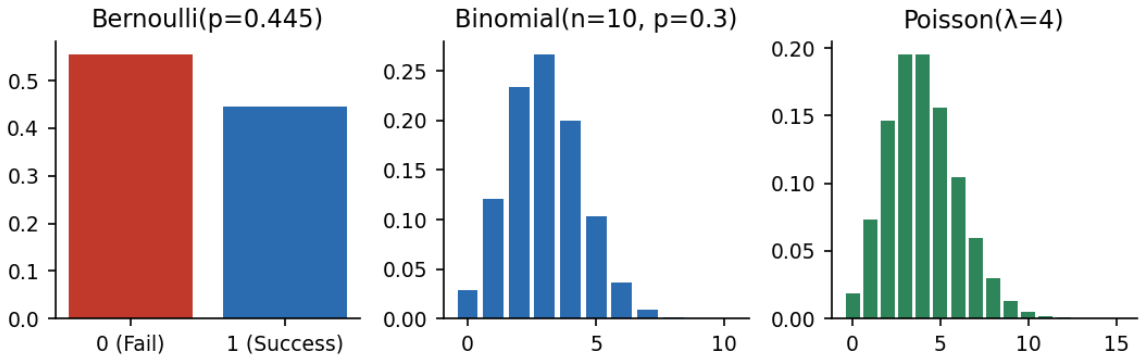
5. Binomial Distribution

Definition. The Binomial distribution gives the **number of successes in n independent Bernoulli trials**, each with the same success probability p.

$$P(X = k) = C(n, k) \cdot p^{k} \cdot (1 - p)^{n-k},\ k = 0, 1, \ldots, n$$

Property	Value
Mean	$n \cdot p$
Variance	$n \cdot p \cdot (1 - p)$ – always smaller than the mean
Special case	Binomial(n = 1, p) = Bernoulli(p)
Conditions	Fixed n · independent trials · constant p · two outcomes

Example. A customer attempts 10 payments, each succeeding with probability 0.3. The number of successes follows Binomial(10, 0.3); the expected number is 10 × 0.3 = 3.



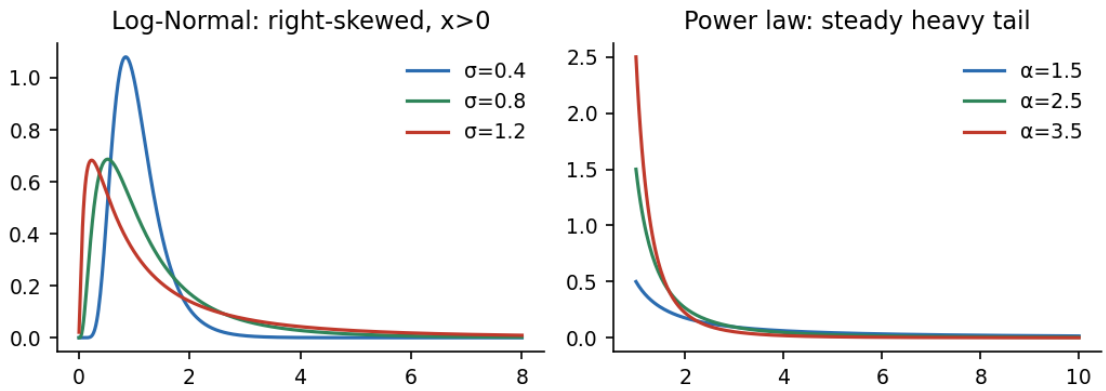
Use in this project. The weekly transaction count is modelled as Binomial with n = 9 (the largest observed count, an assumption) and p = mean/n ≈ 0.317. The data variance (3.23) exceeds the Binomial variance (1.95), so the Binomial fits poorly – the counts are **over-dispersed**.

6. Log-Normal Distribution

Definition. A variable X is Log-Normal if its **natural logarithm ln(X) follows a Normal distribution**. It is defined only for X > 0 and is right-skewed.

$$f(x) = 1 / (x\,\sigma\,\sqrt{(2\pi)}) \cdot \exp(\,-(\ln x - \mu)^2 / (2\sigma^2)\,),\ x > 0$$

Property	Value
Parameters	μ and σ = mean and standard deviation of ln(X)
Median	e^{μ}
Mean	$e^{\mu + \sigma^2/2}$ (larger than the median)
Mode	$e^{\mu - \sigma^2}$ (mode < median < mean)
Origin	Product of many small independent multiplicative effects



Example. Incomes, house prices, stock prices and purchase amounts are typically Log-Normal: most values are moderate, a few are very large.

Use in this project. Fitted to transaction_amount: μ ≈ 8.0, σ ≈ 0.47, median ≈ ₹2,983. It is the best-fitting model (KS p ≈ 0.90).

7. Power Law Distribution

Definition. A power law distribution has a probability that **decreases as a power of the value**: $p(x) \propto x^{-\alpha}$ for $x \geq x_{min}$. Its continuous form is the Pareto distribution.

$$p(x) = ((\alpha - 1) / x_{min}) \cdot (x / x_{min})^{-\alpha},\ x \geq x_{min}$$

$$\text{MLE: } \hat{\alpha} = 1 + n / \Sigma \ln(x_i / x_{min})$$

- **Heavy tail:** very large values are far more likely than under Normal or Log-Normal.
- **Scale-free:** appears as a straight line on a log-log plot of the survival function.
- **"80/20" behaviour:** a few extreme values account for a large share of the total.
- **Usually applies only to the tail** (x above some x_{min}), not the whole data range.
- **Examples:** city populations, wealth, word frequencies, web-page links.

Use in this project. A Pareto model is fitted to the amounts and rejected (KS p ≈ 0): the data have a hump around ₹2,000–4,000 and a lower bound, not a steadily decreasing density from the minimum. Log-Normal is much better.

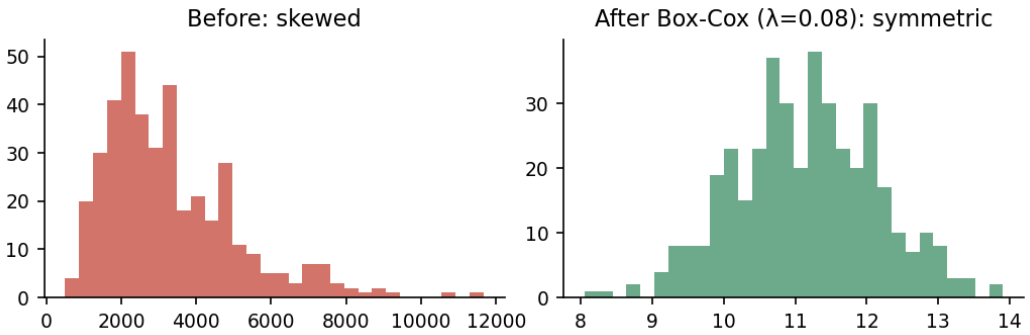
8. Box-Cox Transform

Definition. The Box-Cox transform is a **power transformation that makes positive, skewed data closer to Normal and stabilises its variance**.

$$y(\lambda) = (x^{\lambda} - 1) / \lambda \text{ if } \lambda \neq 0\ ;\ y = \ln(x) \text{ if } \lambda = 0$$

λ	Equivalent transformation
1	No change (data already symmetric)
0.5	Square root
0	Natural logarithm
−1	Reciprocal (1/x)

λ is chosen automatically by **maximum likelihood** so that the transformed data look as normal as possible. A 95% confidence interval for λ that contains 0 means a simple log transform is justified. **Requirement:** all x > 0.



Why it is used. Normality and constant variance are assumptions of regression, ANOVA, t-tests and control charts. After analysis, results can be back-transformed to the original units.

Use in this project. Optimal $\lambda \approx -0.18$ (close to 0). Skewness falls from 3.7 to about 0 and Shapiro-Wilk no longer rejects normality ($p \approx 0.64$).

9. Poisson Distribution

Definition. The Poisson distribution models the **number of events occurring in a fixed interval** of time or space, when events happen independently and at a constant average rate λ.

$$P(X = k) = e^{-\lambda} \cdot \lambda^{k} / k!,\ k = 0, 1, 2, \ldots$$

Property	Value
Mean	λ
Variance	λ (mean = variance)
Conditions	Independent events · constant rate · events cannot occur simultaneously
Check	Dispersion index = variance / mean ≈ 1

Worked example. A call centre receives on average $\lambda = 4$ calls per minute. The probability of exactly 6 calls in a minute is

$$P(X = 6) = e^{-4} \cdot 4^{6} / 6! = 0.0183 \times 4096 / 720 \approx 0.104\ (10.4\%)$$

If variance is much larger than the mean the data are *over-dispersed* and a Negative Binomial model is usually better.

Use in this project. Transactions per day have $\lambda \approx 7.1$ (variance 5.3). The Poisson model is broadly consistent with the data, e.g. P(10 or more transactions in a day) ≈ 18%. It is useful for capacity planning.

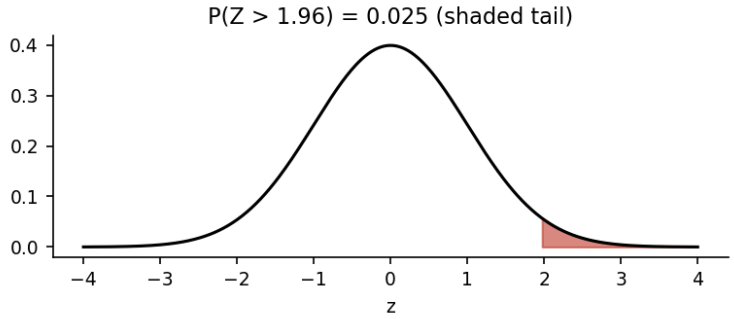
10. Z-score Probability

Definition. A **Z-score** measures how many standard deviations a value lies from the mean. For a Normal variable, Z follows the standard Normal distribution N(0, 1), so **tail probabilities** can be read from the standard Normal CDF Φ.

$$z = (x - \mu) / \sigma$$

$$P(X > x) = 1 - \Phi(z)$$

$$P(X < x) = \Phi(z)$$



- |z| > 2 is unusual; |z| > 3 is typically treated as an outlier.
- Empirical rule for Normal data: 68% within |z| < 1, 95% within |z| < 2, 99.7% within |z| < 3.
- Z-scores put different variables on a common scale so they can be compared.
- **Caution:** the probability is only reliable if the data are approximately Normal. For skewed data, take logs first (Log-Normal Z-score).

Example. Amounts with mean ₹3,365 and σ ₹1,986: for x = ₹5,000, z = (5000 − 3365)/1986 ≈ 0.82, so P(X > 5000) ≈ 20.5% under a Normal model.

Use in this project. The Normal method gives 20.5%, the Log-Normal method 13.9% and the real data 11.4% – showing why the correct distribution matters. Three transactions have |z| > 3.

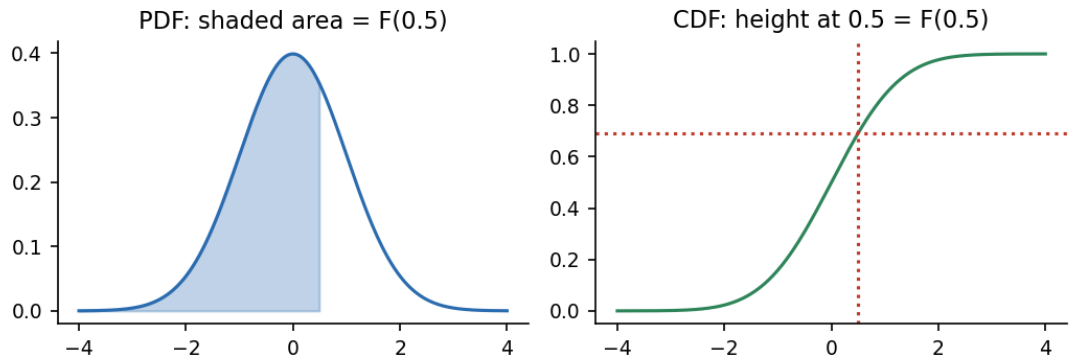
11. PDF vs CDF

PDF (Probability Density Function) f(x) shows the relative likelihood of values; the **area under the curve** between a and b is the probability P(a ≤ X ≤ b). **CDF (Cumulative Distribution Function)** F(x) = P(X ≤ x) gives the **probability of being at or below x**.

$$F(x) = \int f(t)\, dt \text{ (integral from } -\infty \text{ to } x)$$

$$f(x) = dF(x)/dx$$

$$P(a < X \leq b) = F(b) - F(a)$$



	PDF	CDF
Meaning	Density / "shape" of the distribution	Cumulative probability up to x
Range of values	f(x) ≥ 0, can exceed 1	Between 0 and 1
Behaviour	Rises and falls	Never decreases (S-shaped)
Reading probabilities	Area under the curve	Height of the curve (or differences)
Discrete case	PMF: P(X = x)	Step function
Typical use	Spot the mode, skewness and spread	Find percentiles and P(X > x) = 1 − F(x)

Use in this project. The PDF of the amounts is a right-skewed hump (mode ≈ ₹2,400, median ≈ ₹3,000, mean ≈ ₹3,365). The CDF reads ≈ 0.89 at ₹5,000, so about 11% of transactions exceed it.

Appendix A – Glossary

Term	Meaning
Mean (μ)	Average value: sum of values divided by count.
Median	Middle value when sorted; robust to outliers.
Variance (σ^2) / Std. dev. (σ)	Average squared distance from the mean / its square root; measure of spread.
Skewness	Asymmetry of a distribution. 0 = symmetric; > 0 right tail longer.
Kurtosis	Heaviness of tails relative to the Normal (excess kurtosis 0 = Normal).
Quantile / percentile	Value below which a given fraction (%) of the data lie.
PMF / PDF / CDF	Probability mass (discrete) / density (continuous) / cumulative function.
Parameter	A number defining a distribution (μ, σ, p, n, λ, α …).
MLE	Maximum likelihood estimation: choosing parameters that make the observed data most probable.
Over-dispersion	Variance larger than the model expects (e.g. variance > mean for counts).
Outlier	Value unusually far from the rest, e.g. $\|z\| > 3$.
Heavy tail	Extreme values occur with noticeable probability.
Hypothesis test / p-value	A test of a claim (H_0). The p-value is the probability of seeing data this extreme if H_0 were true; $p < 0.05$ usually rejects H_0.
Kolmogorov-Smirnov (KS) test	Compares data with a fitted distribution; $p < 0.05 \Rightarrow$ the model does not fit.
Shapiro-Wilk test	Tests normality; $p < 0.05 \Rightarrow$ not normal.
Chi-square goodness of fit	Compares observed vs expected counts for discrete models.
AIC	Model-comparison score (lower is better), penalises extra parameters.
Empirical CDF	CDF computed directly from the sample data (a step curve).

Appendix B – Formula Cheat Sheet

Distribution	Type	Parameters	Mean	Variance
Bernoulli	Discrete	p	p	$p(1 - p)$
Binomial	Discrete	n, p	np	$np(1 - p)$
Poisson	Discrete	λ	λ	λ
Normal	Continuous	μ, σ	μ	σ^2
Log-Normal	Continuous	μ, σ (of ln X)	$e^{\mu+\sigma^2/2}$	$(e^{\sigma^2} - 1)\, e^{2\mu+\sigma^2}$
Pareto (power law)	Continuous	α, x_{min}	$x_{min}(\alpha-1)/(\alpha-2)$, $\alpha>2$	finite only if $\alpha > 3$

Appendix C – Which Distribution Should I Use?

Situation	Candidate model
One yes/no outcome	Bernoulli
Number of successes in a fixed number of tries	Binomial
Number of events in a time/space interval	Poisson (Negative Binomial if over-dispersed)
Symmetric, bell-shaped measurement	Normal
Positive, right-skewed, multiplicative effects (money, size)	Log-Normal
Extremely heavy tail, straight on log-log plot	Power law / Pareto (tail only)
Skewed positive data you must make normal	Box-Cox (or log) transform first